IV. Preface

In an era of rapid globalization and evolving trade dynamics, understanding and forecasting export trends is crucial for any nation's economic strategy. India, with its diverse and burgeoning economy, presents a unique case study in export performance and potential. This paper aims to provide a comprehensive methodology for forecasting India's export trends and analyzing historical export data from 1999 to 2023, encapsulating insights that can guide policymakers, economists, and business leaders.

The paper is divided into two main components: Export Forecasting and Export Analysis and Visualization. In the Export Forecasting section, we meticulously split the data into training (1999-2019) and testing (2020-2023) sets, identifying the Seasonal Autoregressive Integrated Moving Average (SARIMA) model as the top performer. This model is employed to forecast export values from 2024 to 2029, offering a forward-looking perspective grounded in robust statistical analysis.

The Export Analysis and Visualization section delves into export data collected from 2010 to 2021. This data is systematically grouped by commodity category and trade region, then presented through an interactive Power BI dashboard. This visualization tool not only highlights significant export trends but also showcases India's dynamic position in the global market. Our analysis reveals that while India's recent export surges underscore its economic resilience, there remain ample opportunities to refine forecasting methodologies and enhance the dashboard's capabilities for more effective decision-making in international trade.